



Qualification Pack

Transformer Manufacturing Engineer

QP Code: CUTM/PSS/Q0103

Version: 1.0

NSQF Level: 6

Centurion University of Technology and Management || Aluri Nagar, Paralakhemundi, R Sitapur
Uppalada || email:partha@cutm.ac.in



Qualification Pack

Contents

CUTM/PSS/Q0103: Transformer Manufacturing Engineer	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CUTM/PSS/N0105: Review transformer manufacturing requirements	5
CUTM/PSS/N0106: Perform transformer manufacturing Process	10
CUTM/PSS/N0107: Review Transformer Testing requirements	17
CUTM/PSS/N0108: Testing and quality check of assembled transformer.	22
CUTM/VSQ/N7801: Ensure organizational health, safety, and security	27
DGT/VSQ/N0103: Employability Skills (90 Hours)	32
Assessment Guidelines and Weightage	40
<i>Assessment Guidelines</i>	40
<i>Assessment Weightage</i>	42
Acronyms	43
Glossary	44

Qualification Pack

CUTM/PSS/Q0103: Transformer Manufacturing Engineer

Brief Job Description

The Individual is responsible to review the customer requirement, transformer design documents, verification of procured material, and supporting the manufacturing team for quality transformer fabrication along with testing transformers to meet customer requirements.

Personal Attributes

A Transformer Manufacturing Engineer should be detail-oriented, safety-conscious, and technically proficient. They must possess strong analytical thinking, problem-solving ability, and effective communication skills. A commitment to quality, teamwork, time management, and adaptability to emerging technologies is essential. Integrity and a proactive mindset are key to success in this role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/PSS/N0105: Review transformer manufacturing requirements](#)
2. [CUTM/PSS/N0106: Perform transformer manufacturing Process](#)
3. [CUTM/PSS/N0107: Review Transformer Testing requirements](#)
4. [CUTM/PSS/N0108: Testing and quality check of assembled transformer.](#)
5. [CUTM/VSQ/N7801: Ensure organizational health, safety, and security](#)
6. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Power Distribution
Occupation	Transformer Manufacturing
Country	India
NSQF Level	6
Credits	20

Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	B.E./B.Tech (in Mechanical / Electrical / Electronics / Production /Mechatronics) with NA of experience OR Pursuing 4th year UG (in case of 4-year UG with honours/ honours with research) (in Mechanical / Electrical / Electronics / Production /Mechatronics) with NA of experience OR Completed 2nd year diploma after 12th (in Mechanical / Electrical / Production / Electronics / Mechatronics) with 2 Years of experience OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience in relevant field OR Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience in relevant field
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	22/06/2026
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	QG-06-PW-00617-2023-V1-CUTM
NQR Version	1.0

Qualification Pack

CUTM/PSS/N0105: Review transformer manufacturing requirements

Description

It covers the skills and knowledge required to interpret and review customer specifications, technical documents, and applicable standards for transformer manufacturing. It involves assessing manufacturing feasibility, preparing documentation, and coordinating with relevant teams to ensure alignment with product design, quality requirements, and organizational capabilities.

Scope

The scope covers the following :

- It covers the competencies required to evaluate customer and project-specific requirements related to transformer manufacturing. It includes interpreting technical documentation, standards, and specifications to identify design, raw material, and process needs, ensuring that all manufacturing requirements align with regulatory, safety, and quality standards. The scope also involves coordinating with internal teams and stakeholders to facilitate efficient planning and execution of transformer production.

Elements and Performance Criteria

Review the customer requirement, engineering drawing and standards

To be competent, the user/individual on the job must be able to:

- PC1.** interact with the head of transformer manufacturing on customer requirement, engineering drawings and standards to be followed for transformer manufacturing process.
- PC2.** Review tools, raw materials and consumables required for production in sufficient quantity as per production plan/operators' instructions.
- PC3.** clearly check the application of transformer and standards to be followed for manufacturing and testing process.

Reading of detailed assembly drawing, dimensional tolerances, and assembly process

To be competent, the user/individual on the job must be able to:

- PC4.** Review detailed assembly drawing, material grade, dimensional tolerances, and assembly process for the specific order to be executed.
- PC5.** distribute the assembly drawing, standards, and process steps to be followed for each stage.
- PC6.** distribute the work order details required for each process, batch details to be followed for the manufacturing stage.
- PC7.** get necessary clarification from the appropriate authorities in case of any doubts.

Review all bill of materials

To be competent, the user/individual on the job must be able to:

- PC8.** Review all bill of materials along with quantity required transformer manufacturing process with the expected quality requirements.
- PC9.** ensure that required raw material/tools is procured from stores /vendor before starting work.
- PC10.** check incoming quality Inspection reports on meeting the quality and quantity requirements.

Qualification Pack

- PC11.** ensure tools required are placed in the respective stages of transformer manufacturing process.
- PC12.** if tool is not available collect the tools from stores / vendor site and place it in the respective stages.
- PC13.** update all the material transaction details in appropriate logs, template, MIS.
- PC14.** maintain production file, log, template MIS and update as per requirement.
- PC15.** report to Head of manufacturing in case of any dispute in the drawing and standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company policies and procedures related to transformer design and manufacturing.
- KU2.** Roles and responsibilities within the transformer design/manufacturing team.
- KU3.** Compliance with relevant codes, standards, and regulations.
- KU4.** Types of transformers (e.g., power, distribution, dry-type) and their industrial applications.
- KU5.** Key design parameters and specifications (electrical & mechanical).
- KU6.** Materials used for core, winding, insulation, and tank components.
- KU7.** Ability to interpret technical drawings, BOMs, and datasheets.
- KU8.** Understanding of core processes: core building, winding, insulation, assembly, and testing.
- KU9.** Familiarity with customer-specific technical requirements and design expectations.
- KU10.** Understanding of cost, material availability, and manufacturability considerations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret technical documents
- GS2.** Identify materials, components, and tools
- GS3.** Communicate clearly and effectively
- GS4.** Document reports, plans, and analysis
- GS5.** Think critically and analytically
- GS6.** Practice professional conduct, ethics, and responsibility
- GS7.** Maintain records and documentation
- GS8.** Use digital tools and applications for reporting, scheduling, and document sharing.
- GS9.** Apply problem-solving skills

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Review the customer requirement, engineering drawing and standards</i>	6	18	-	-
PC1. interact with the head of transformer manufacturing on customer requirement, engineering drawings and standards to be followed for transformer manufacturing process.	2	6	-	-
PC2. Review tools, raw materials and consumables required for production in sufficient quantity as per production plan/operators' instructions.	2	6	-	-
PC3. clearly check the application of transformer and standards to be followed for manufacturing and testing process.	2	6	-	-
<i>Reading of detailed assembly drawing, dimensional tolerances, and assembly process</i>	8	11	-	-
PC4. Review detailed assembly drawing, material grade, dimensional tolerances, and assembly process for the specific order to be executed.	2	5	-	-
PC5. distribute the assembly drawing, standards, and process steps to be followed for each stage.	2	-	-	-
PC6. distribute the work order details required for each process, batch details to be followed for the manufacturing stage.	2	2	-	-
PC7. get necessary clarification from the appropriate authorities in case of any doubts.	2	4	-	-
<i>Review all bill of materials</i>	16	41	-	-
PC8. Review all bill of materials along with quantity required transformer manufacturing process with the expected quality requirements.	2	3	-	-
PC9. ensure that required raw material/tools is procured from stores /vendor before starting work.	2	6	-	-
PC10. check incoming quality Inspection reports on meeting the quality and quantity requirements.	2	6	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure tools required are placed in the respective stages of transformer manufacturing process.	2	6	-	-
PC12. if tool is not available collect the tools from stores / vendor site and place it in the respective stages.	2	6	-	-
PC13. update all the material transaction details in appropriate logs, template, MIS.	2	6	-	-
PC14. maintain production file, log, template MIS and update as per requirement.	2	3	-	-
PC15. report to Head of manufacturing in case of any dispute in the drawing and standards.	2	5	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0105
NOS Name	Review transformer manufacturing requirements
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQF Clearance Date	23/06/2023

Qualification Pack

CUTM/PSS/N0106: Perform transformer manufacturing Process

Description

This NOS involves supervising and performing the complete transformer manufacturing process, including core assembly, winding, and final assembly. It ensures adherence to safety, quality standards, and technical specifications while coordinating with other departments. It also emphasizes resource planning, process inspection, troubleshooting, and maintaining proper documentation for quality assurance.

Scope

The scope covers the following :

- Overseeing and executing the complete transformer manufacturing workflow, including coil winding, core assembly, tank fabrication, insulation, oil processing, and testing, while ensuring quality standards, safety compliance, use of tools and materials, process optimization, and coordination with design and production teams.

Elements and Performance Criteria

Confirm availability of required quantity as per transformer design

To be competent, the user/individual on the job must be able to:

- PC1.** check quantity available at each process stage as per transformer design.
- PC2.** report the concern in case of any quality / quantity discrepancy.
- PC3.** arrange for discrepant quantity/quality of materials.

Core Assembly

To be competent, the user/individual on the job must be able to:

- PC4.** understand core size, material, and grain orientation.
- PC5.** cut core shape as per design.
- PC6.** understand different core stacking methods.
- PC7.** use of temporary bolt guides for stacking.
- PC8.** stake core as per assembly drawing requirement with appropriate quantity.
- PC9.** tighten bolt to avoid gap between cores.
- PC10.** insulate assembled core.

Coil Assembly

To be competent, the user/individual on the job must be able to:

- PC11.** understand coil size, material and number of coils as per design.
- PC12.** keep winding room separate from other manufacturing areas with positive pressure and stake horizontal/vertical winding mandrels.
- PC13.** check conductor is either copper magnetic wire or continuously transposed conductor.
- PC14.** check conductor purchased are pre-wrapped with thermally upgraded paper or nomex.
- PC15.** check high strength wire or epoxy bonded CTC used when high short circuit forces required as per design requirement.

Qualification Pack

- PC16.** chose winding type according to voltage & application, cont. disc and design interleaved disc, helical, etc.
- PC17.** make coil windings as per assembly requirement.
- PC18.** complete lead connection with stress ring and pick tail connected with lead.
- PC19.** complete winding insulation package.
- PC20.** complete lead braising

Coil & Core Assembly

To be competent, the user/individual on the job must be able to:

- PC21.** bond exposed edges of core with low viscosity, high strength epoxy resin which penetrates and bonds laminations.
- PC22.** clamp prefabricated coil with insulations placed on bottom clamps.
- PC23.** clamp windings using external or internal tie rods to provide additional support for axial forces.
- PC24.** rigidly support leads and busbars to withstand forces from shipping and short circuits.
- PC25.** move assembly on air cushion.
- PC26.** make necessary LTC / DTC lead connection
- PC27.** link board with re-connectable LV.

Vapor phase unit

To be competent, the user/individual on the job must be able to:

- PC28.** complete core and coil assembly is dried using a vapor phase cycle method.
- PC29.** maintain required oven temperature and time for curing.
- PC30.** continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.

Oil Tank fabrication

To be competent, the user/individual on the job must be able to:

- PC31.** understand tank requirements and study tank assembly drawing.
- PC32.** fabricate tank with appropriate process steps and materials.
- PC33.** maintain required oven temperature and time for curing.
- PC34.** continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.

Transformer Assembly

To be competent, the user/individual on the job must be able to:

- PC35.** understand tank requirements and study tank assembly drawing.
- PC36.** fabricate tank with appropriate process steps and materials.
- PC37.** make grit-blasted cleaning the tank before priming and painting.
- PC38.** provide necessary LTC connections.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Importance of adhering to organizational procedures, policies, and SOPs during the manufacturing process.
- KU2.** health, safety, and environmental guidelines as per the company's practices.
- KU3.** Types of transformers (e.g., power, distribution, dry-type) and their specific manufacturing requirements.
- KU4.** Core materials, winding conductors (copper/aluminum), insulation materials, and their properties.
- KU5.** Use and operation of tools, jigs, fixtures, and machines involved in the process (e.g., coil winding machines, core building machines).
- KU6.** Tolerances and specifications based on transformer design drawings.
- KU7.** Importance of maintaining cleanliness, material traceability, and avoiding contamination.
- KU8.** Preventive maintenance techniques and scheduling for machinery used in manufacturing.
- KU9.** National and international standards applicable to transformer manufacturing (e.g., IS, IEC).
- KU10.** Safe handling of heavy equipment, sharp tools, flammable insulation materials, and transformer oil.
- KU11.** Usage of Personal Protective Equipment (PPE) during operations.
- KU12.** Emergency procedures in case of fire, electrical hazards, or equipment failure.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Coordinate with team members, supervisors, and other departments to ensure seamless operations.
- GS2.** Apply reasoning and problem-solving skills to address manufacturing issues efficiently.
- GS3.** Operate digital tools and software used in transformer design, control systems, and documentation.
- GS4.** Read and interpret engineering drawings, SOPs, and component specifications accurately.
- GS5.** Prioritize tasks and manage time to meet production goals within deadlines.
- GS6.** Maintain precise records of manufacturing, inspection, and quality control.
- GS7.** Work effectively in teams and contribute to process improvements.
- GS8.** Follow safety guidelines, quality standards, and environmental regulations diligently.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Confirm availability of required quantity as per transformer design</i>	6	9	-	-
PC1. check quantity available at each process stage as per transformer design.	2	3	-	-
PC2. report the concern in case of any quality / quantity discrepancy.	2	3	-	-
PC3. arrange for discrepant quantity/quality of materials.	2	3	-	-
<i>Core Assembly</i>	14	19	-	-
PC4. understand core size, material, and grain orientation.	2	3	-	-
PC5. cut core shape as per design.	2	3	-	-
PC6. understand different core stacking methods.	2	3	-	-
PC7. use of temporary bolt guides for stacking.	2	3	-	-
PC8. stake core as per assembly drawing requirement with appropriate quantity.	2	3	-	-
PC9. tighten bolt to avoid gap between cores.	2	1	-	-
PC10. insulate assembled core.	2	3	-	-
<i>Coil Assembly</i>	20	32	-	-
PC11. understand coil size, material and number of coils as per design.	2	3	-	-
PC12. keep winding room separate from other manufacturing areas with positive pressure and stake horizontal/vertical winding mandrels.	2	3	-	-
PC13. check conductor is either copper magnetic wire or continuously transposed conductor.	1	2	-	-
PC14. check conductor purchased are pre-wrapped with thermally upgraded paper or nomex.	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. check high strength wire or epoxy bonded CTC used when high short circuit forces required as per design requirement.	2	3	-	-
PC16. chose winding type according to voltage & application, cont. disc and design interleaved disc, helical, etc.	2	3	-	-
PC17. make coil windings as per assembly requirement.	3	6	-	-
PC18. complete lead connection with stress ring and pick tail connected with lead.	2	3	-	-
PC19. complete winding insulation package.	2	3	-	-
PC20. complete lead braising	2	3	-	-
<i>Coil & Core Assembly</i>	15	24	-	-
PC21. bond exposed edges of core with low viscosity, high strength epoxy resin which penetrates and bonds laminations.	2	3	-	-
PC22. clamp prefabricated coil with insulations placed on bottom clamps.	2	3	-	-
PC23. clamp windings using external or internal tie rods to provide additional support for axial forces.	3	6	-	-
PC24. rigidly support leads and busbars to withstand forces from shipping and short circuits.	2	3	-	-
PC25. move assembly on air cushion.	2	3	-	-
PC26. make necessary LTC / DTC lead connection	2	3	-	-
PC27. link board with re-connectable LV.	2	3	-	-
<i>Vapor phase unit</i>	8	12	-	-
PC28. complete core and coil assembly is dried using a vapor phase cycle method.	2	3	-	-
PC29. maintain required oven temperature and time for curing.	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.	4	6	-	-
<i>Oil Tank fabrication</i>	10	15	-	-
PC31. understand tank requirements and study tank assembly drawing.	2	3	-	-
PC32. fabricate tank with appropriate process steps and materials.	2	3	-	-
PC33. maintain required oven temperature and time for curing.	4	6	-	-
PC34. continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.	2	3	-	-
<i>Transformer Assembly</i>	7	9	-	-
PC35. understand tank requirements and study tank assembly drawing.	2	3	-	-
PC36. fabricate tank with appropriate process steps and materials.	2	3	-	-
PC37. make grit-blasted cleaning the tank before priming and painting.	2	2	-	-
PC38. provide necessary LTC connections.	1	1	-	-
NOS Total	80	120	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0106
NOS Name	Perform transformer manufacturing Process
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

CUTM/PSS/N0107: Review Transformer Testing requirements

Description

This NOS involves reviewing transformer testing requirements by analyzing design specifications and standards, planning and coordinating testing activities, ensuring correct setup and safety, supervising test execution, analyzing results, and preparing reports. It ensures compliance with national and international standards, using calibrated instruments and collaboration across production and quality teams.

Scope

The scope covers the following :

- The scope of this NOS includes evaluating customer specifications, interpreting technical drawings and testing standards, planning test setups, ensuring compliance with quality and safety norms, coordinating with design and testing teams, and documenting results for effective product validation and delivery.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** interact with Transformer Design Engineer on customer requirement, engineering drawings and standards to be followed.
- PC2.** Review test tools and consumables used for manufacturing.
- PC3.** clearly understand the application of transformer and standards to be followed for testing process.
- PC4.** understand detailed specification, material grade, dimensional tolerances, and assembly process for the specific batch to be tested.
- PC5.** Review the process steps followed during manufacturing.
- PC6.** Verify the work sheet steps on each stage of process followed matching to Client requirement.
- PC7.** get necessary clarification from the appropriate authorities in case of any doubts.
- PC8.** Verify the working condition of all Testing equipment.
- PC9.** Check the calibration validity of the measuring instruments to be used for Testing.
- PC10.** check incoming quality inspection reports on meeting the quality and quantity requirements.
- PC11.** ensure tools required are placed in respective stages of transformer testing.
- PC12.** if tool is unavailable collect tools from stores / vendor site and place it in respective stage.
- PC13.** update all testing details in appropriate logs, template, MIS.
- PC14.** maintain test file, log, template, MIS and update as per requirement.
- PC15.** report to Head of manufacturing in case of any dispute in drawing, standards, materials, consumables, tools, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Importance of quality assurance protocols in transformer manufacturing.
- KU2.** Safety and environmental policies related to electrical testing.
- KU3.** Documentation and reporting procedures within the organization.
- KU4.** Coordination with procurement, design, and production teams.
- KU5.** Principles of transformer operation and testing requirements.
- KU6.** National and international standards for transformer testing.
- KU7.** Electrical parameters: voltage ratio, winding resistance, insulation resistance, vector group, short circuit impedance, etc.
- KU8.** Use of testing equipment like megger, turns ratio meter, high voltage tester, etc.
- KU9.** Calibration and maintenance of testing instruments.
- KU10.** Basics of electrical measurements and instrumentation.
- KU11.** Preventive and corrective action planning.
- KU12.** Basics of data analysis and use of software tools for test data recording.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret transformer testing standards and ensure compliance with customer and regulatory requirements.
- GS2.** Analyze test results, identify deviations, and recommend corrective or preventive actions.
- GS3.** Communicate effectively with testing teams, design, and quality departments.
- GS4.** Coordinate testing schedules in alignment with production timelines.
- GS5.** Prepare, review, and maintain accurate test reports and documentation.
- GS6.** Utilize digital tools and ERP systems for test planning and data analysis.
- GS7.** Make informed decisions based on test data to approve or reject transformers.
- GS8.** Stay updated on industry testing standards and apply them in review processes.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. interact with Transformer Design Engineer on customer requirement, engineering drawings and standards to be followed.	2	5	-	-
PC2. Review test tools and consumables used for manufacturing.	2	-	-	-
PC3. clearly understand the application of transformer and standards to be followed for testing process.	2	5	-	-
PC4. understand detailed specification, material grade, dimensional tolerances, and assembly process for the specific batch to be tested.	2	5	-	-
PC5. Review the process steps followed during manufacturing.	2	5	-	-
PC6. Verify the work sheet steps on each stage of process followed matching to Client requirement.	2	5	-	-
PC7. get necessary clarification from the appropriate authorities in case of any doubts.	2	5	-	-
PC8. Verify the working condition of all Testing equipment.	2	5	-	-
PC9. Check the calibration validity of the measuring instruments to be used for Testing.	2	5	-	-
PC10. check incoming quality inspection reports on meeting the quality and quantity requirements.	2	5	-	-
PC11. ensure tools required are placed in respective stages of transformer testing.	2	5	-	-
PC12. if tool is unavailable collect tools from stores / vendor site and place it in respective stage.	2	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. update all testing details in appropriate logs, template, MIS.	2	5	-	-
PC14. maintain test file, log, template, MIS and update as per requirement.	2	5	-	-
PC15. report to Head of manufacturing in case of any dispute in drawing, standards, materials, consumables, tools, etc	2	5	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0107
NOS Name	Review Transformer Testing requirements
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

CUTM/PSS/N0108: Testing and quality check of assembled transformer.

Description

This NOS covers testing and quality checks of assembled transformers to ensure performance, safety, and compliance with standards. It includes conducting electrical tests, verifying insulation, assessing load performance, identifying defects, and analyzing test reports to certify transformer reliability in power distribution and industrial applications.

Scope

The scope covers the following :

- This NOS includes conducting electrical tests, insulation checks, load performance assessments, and verifying compliance with standards. It covers fault detection, quality assurance, report analysis, and ensuring transformer reliability. It ensures assembled transformers meet operational, safety, and industry standards for efficient power distribution and industrial applications.

Elements and Performance Criteria

Various testing methods as per transformer design and applications

To be competent, the user/individual on the job must be able to:

- PC1.** understand various testing methods to be followed as per design and application.
- PC2.** prepare check list of parameters for consideration during testing and inspection.
- PC3.** verify connections of HT/LT side and inspect bolt/lugs and solder of electrical connections.
- PC4.** perform routine test, loss measurement and temperature rise test, dielectric test.
- PC5.** carry out insulation-resistance test to identify grounding and shorting faults in the connections.
- PC6.** check for any damage, flash, and hair crack with respect to specified parameters.
Parameters: condition of OLTC, leakage of oil in bushing collar, porcelain insulator bushing, gasket joints and flanges.
- PC7.** inspect excess humming noise in transformer and identify loose fitting of silicon mixed steel alloys laminated core joints.

Quality check and confirmation of proper assembly

To be competent, the user/individual on the job must be able to:

- PC8.**
 - identify faults by visually inspecting general appearance, leakage of oil, and physical condition
 - Physical condition: rust on body and radiators
- PC9.** check silica gel breather for the oil level in oil cap on specific parameters.
- PC10.**
 - identify faults
 - Faults: faults due to primary winding, braze /solder of LT winding
 - joints melted, overheat, open circuit in internal wiring etc.
- PC11.**
 - maintain records of transformer basic information
 - Basic information: serial number, diagram, rating plate etc. and defects/repair showing diagnostic records for assessing the transformer performance history

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the transformer principles, electromagnetic induction, and types of transformers.
- KU2.** Various transformer parts such as core, windings, insulation, bushings, tap changers, and cooling systems.
- KU3.** the importance of visual inspection, insulation checks, and ensuring proper assembly before testing.
- KU4.** The key transformer tests like turns ratio test, insulation resistance test, dielectric strength test, and winding resistance test.
- KU5.** Awareness of mechanical stress testing, oil testing (BDV test), and temperature rise tests for quality assurance.
- KU6.** different types of defects such as insulation failure, short circuits, overheating, and improper connections.
- KU7.** safety measures, PPE usage, grounding techniques, and emergency handling in transformer testing.
- KU8.** to record test results, prepare quality check reports, and ensure compliance with specifications.
- KU9.** Ensuring transformers meet customer requirements, efficiency targets, and industry quality standards before dispatch.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to clearly explain test results, quality standards, and troubleshooting steps to team members and supervisors.
- GS2.** Identifying transformer defects, analyzing test data, and determining corrective actions.
- GS3.** Ensuring accuracy in test procedures, data recording, and compliance with quality standards.
- GS4.** Managing multiple quality checks and tests within project deadlines.
- GS5.** Following electrical safety guidelines, PPE usage, and emergency protocols in the testing environment.
- GS6.** test insulation resistance, turns ratio, short circuit, and oil breakdown voltage tests.
- GS7.** Analyze and validate test results against industry standards and manufacturer specifications.
- GS8.** Proficiency in handling megger, TTR meter, high-voltage testers, and infrared thermography tools.
- GS9.** Preparing quality check reports, maintaining records, and documenting non-conformities for corrective action.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Various testing methods as per transformer design and applications</i>	26	39	-	-
PC1. understand various testing methods to be followed as per design and application.	4	6	-	-
PC2. prepare check list of parameters for consideration during testing and inspection.	4	6	-	-
PC3. verify connections of HT/LT side and inspect bolt/lugs and solder of electrical connections.	2	3	-	-
PC4. perform routine test, loss measurement and temperature rise test, dielectric test.	4	6	-	-
PC5. carry out insulation-resistance test to identify grounding and shorting faults in the connections.	4	6	-	-
PC6. check for any damage, flash, and hair crack with respect to specified parameters. Parameters: condition of OLTC, leakage of oil in bushing collar, porcelain insulator bushing, gasket joints and flanges.	4	6	-	-
PC7. inspect excess humming noise in transformer and identify loose fitting of silicon mixed steel alloys laminated core joints.	2	6	-	-
<i>Quality check and confirmation of proper assembly</i>	14	21	-	-
PC8. • identify faults by visually inspecting general appearance, leakage of oil, and physical condition • Physical condition: rust on body and radiators	4	6	-	-
PC9. check silica gel breather for the oil level in oil cap on specific parameters.	2	3	-	-
PC10. • identify faults • Faults: faults due to primary winding, braze /solder of LT winding • joints melted, overheat, open circuit in internal wiring etc.	4	6	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. - maintain records of transformer basic information - Basic information: serial number, diagram, rating plate etc. and defects/repair showing diagnostic records for assessing the transformer performance history	4	6	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0108
NOS Name	Testing and quality check of assembled transformer.
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

CUTM/VSQ/N7801: Ensure organizational health, safety, and security

Description

This NOS focuses on maintaining workplace safety, health, and security by identifying hazards, following safety procedures, using personal protective equipment (PPE), and responding to emergencies. It ensures compliance with health and safety regulations, proper waste disposal, and adherence to security protocols to create a safe and productive work environment.

Scope

The scope covers the following :

- This NOS covers workplace hazard identification, adherence to safety regulations, proper use of personal protective equipment (PPE), emergency response, fire safety, first aid, and waste management. It also includes implementing security measures, ensuring workplace hygiene, and promoting a safety-conscious culture to prevent accidents and maintain a secure working environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.
- PC2.** recognize other possible security issues existing in the workplace.
- PC3.** plan safety techniques for self and others.
- PC4.** recognize different measures to curb the hazards.
- PC5.** implement programs for health, safety and security of the people at the workplace.
- PC6.** communicate the safety plans to everyone at the workplace.
- PC7.** attach disciplinary rules with the implementation at the workplace.
- PC8.** comply with regulatory health and safety guidelines applicable to the workplace.
- PC9.** use and maintain protective equipment as per requirement in a production house.
- PC10.** ensure organizational activities in line with approved guidelines and procedures.
- PC11.** undertake and advise healthy lifestyle for self and others.
- PC12.** follow environment management system related procedures.
- PC13.** ensure safe handling and disposal of waste and debris.
- PC14.** minimize health and safety risks to self and others due to own actions.
- PC15.** monitor the workplace and work processes for potential risks and threats to self and others.
- PC16.** carry out periodic check to keep work area free from hazards and obstructions.
- PC17.** ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.
- PC18.** organize first aid, firefighting and emergency training sessions for others.
- PC19.** instruct and guide others to take action in the event of fire, emergencies or accidents.

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines, workplace policies, and compliance with national and industry safety standards.
- KU2.** common hazards such as electrical risks, fire, chemical exposure, and physical injuries.
- KU3.** different types of PPE (gloves, helmets, safety shoes, goggles) and their correct usage.
- KU4.** the proper handling, operation, and maintenance of workplace tools to prevent accidents.
- KU5.** Awareness of first aid procedures, fire evacuation plans, and emergency contact protocols.
- KU6.** fire hazards, fire extinguisher types, and safe fire-handling procedures in case of an incident.
- KU7.** the importance of keeping the workplace clean, organized, and free of potential hazards.
- KU8.** proper lifting posture, use of lifting tools, and avoiding strain-related injuries.
- KU9.** workplace security protocols, restricted areas, and unauthorized access prevention.
- KU10.** safe disposal methods for hazardous and non-hazardous materials to ensure environmental compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** report safety concerns, follow instructions, and communicate hazards to supervisors and colleagues.
- GS2.** Identify workplace safety issues and taking appropriate actions to prevent accidents.
- GS3.** Handle safety-related measurements like voltage limits, weight capacity, and fire extinguisher ratings.
- GS4.** Ensure timely completion of tasks while following safety protocols and risk management procedures.
- GS5.** Work effectively with colleagues to maintain a safe and secure work environment.
- GS6.** Identify risks such as electrical shocks, fire, slippery surfaces, and chemical exposure in a workplace.
- GS7.** Ensure the safe use of machines, hand tools, and electrical equipment to prevent accidents.
- GS8.** Follow safety protocols, emergency procedures, and workplace safety policies.
- GS9.** do basic first aid techniques, fire safety procedures, and responding to workplace accidents.
- GS10.** Ensure proper disposal of hazardous and non-hazardous waste in compliance with environmental standards.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.	2	4	-	-
PC2. recognize other possible security issues existing in the workplace.	2	4	-	-
PC3. plan safety techniques for self and others.	2	4	-	-
PC4. recognize different measures to curb the hazards.	2	4	-	-
PC5. implement programs for health, safety and security of the people at the workplace.	2	4	-	-
PC6. communicate the safety plans to everyone at the workplace.	2	4	-	-
PC7. attach disciplinary rules with the implementation at the workplace.	2	4	-	-
PC8. comply with regulatory health and safety guidelines applicable to the workplace.	2	4	-	-
PC9. use and maintain protective equipment as per requirement in a production house.	2	3	-	-
PC10. ensure organizational activities in line with approved guidelines and procedures.	2	3	-	-
PC11. undertake and advise healthy lifestyle for self and others.	1	3	-	-
PC12. follow environment management system related procedures.	1	3	-	-
PC13. ensure safe handling and disposal of waste and debris.	2	4	-	-
PC14. minimize health and safety risks to self and others due to own actions.	1	4	-	-
PC15. monitor the workplace and work processes for potential risks and threats to self and others.	1	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. carry out periodic check to keep work area free from hazards and obstructions.	1	4	-	-
PC17. ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.	1	4	-	-
PC18. organize first aid, firefighting and emergency training sessions for others.	1	3	-	-
PC19. instruct and guide others to take action in the event of fire, emergencies or accidents.	1	3	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7801
NOS Name	Ensure organizational health, safety, and security
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

Qualification Pack

- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

Qualification Pack

- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

Qualification Pack

- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Strategy

1. Assessment System Overview:

- Batches would be assigned to Assessment Agencies as per the sectors for conducting assessment on CUTM portal or by email.
- Assessment agencies will have to give a confirmation regarding the assigned batches and share the particular Question set to Centurion University.
- Assessment agency deploys the ToA certified Assessor for conducting the assessment from the pool of verified assessor by Centurion University.
- Centurion University will monitor the assessment process and keep the records online or offline (wherever needed)

2. Procedure at Testing Centre:



Qualification Pack

- Check the Assessment location, date and time
- Maximum batch size will be of 30.
- Assessor will Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.
- Assessor will ensure all the equipment's and required materials are available at the testing centre.

3. Assessment Quality Assurance:

- Question bank is created by the Subject Matter Experts (SME) of Assessment Agencies verified by Centurion University.
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified
- Surprise visit would be done time to time.

4. Assessment Evidence collection:

- Geo tagged photographs of assessment sessions would be captured by the Assessor.
- Video call verification will be done by Centurion University during assessment and keep the recorded videos.

5. Minimum Passing Criteria:

- 70% overall to pass.
- Minimum 70% in each NOS

6. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

7. Assessor Details: ToA Certified Assessor with relevant experience

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/PSS/N0105.Review transformer manufacturing requirements	30	70	-	-	100	20
CUTM/PSS/N0106.Perform transformer manufacturing Process	80	120	-	-	200	30
CUTM/PSS/N0107.Review Transformer Testing requirements	30	70	-	-	100	15
CUTM/PSS/N0108.Testing and quality check of assembled transformer.	40	60	-	-	100	15
CUTM/VSQ/N7801.Ensure organizational health, safety, and security	30	70	0	0	100	10
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	230	420	0	0	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.